

单词统计器

复习

- | | |
|----------------|----|
| 1. 数据流的概念 | 了解 |
| 2. 文件处理简介 | 了解 |
| 3. 文件的输入与输出 | 重点 |
| 4. 文件写入案例 | 熟悉 |
| 5. 文件读取案例 | 熟悉 |
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6. main函数调用

案例展示

```
data.txt
1 i love china i love beijing
```

Microsoft Visual Studio 调试控制台

```
i love china i love beijing
```

```
单词集合为:
```

```
beijing china i love
```

```
单词集合及出现次数:
```

```
beijing:出现1次
```

```
china:出现1次
```

```
i:出现2次
```

```
love:出现2次
```

```
E:\UStudioCode\Test001\x64\Debug\Test001.exe
```

```
按任意键关闭此窗口. . .
```

创建word单词基础类

添加->类

添加类

类名(L) .h 文件(F) .cpp 文件(P)

1.写上类名

基类(B) 访问(A)

 public

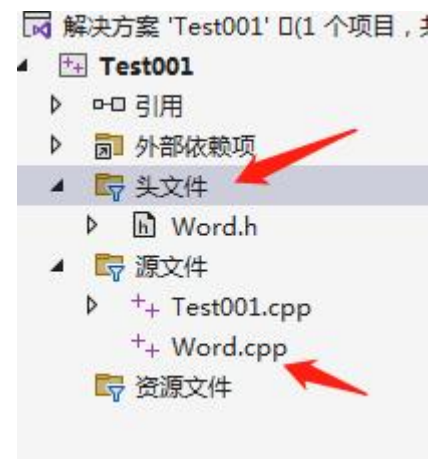
其他选项:

☐ 内联(I)

☐ 托管(M)

2.点击确定

确定 取消



Word.h

```
#pragma once
#include<string>
using namespace std;
class Word
{
public:
    Word(string s);
    string getWord()const;
    bool operator<(const Word& a) const;
    bool operator==(string s);
private:
    string s;
};
```

Word.cpp

```
#include "Word.h"
```

```
Word::Word(string s) {
```

```
    this->s = s;
```

```
}
```

```
string Word::getWord() const{
```

```
    return s;
```

```
}
```

```
bool Word::operator<(const Word& a) const{
```

```
    return s < a.getWord();
```

```
}
```

```
bool Word::operator==(string s) {
```

```
    return this->s == s;
```

```
}
```

创建WordSet集合类

步骤和Word类一样。

WordSet.h

```
#pragma once
#include <iostream>
#include "Word.h"
#include <set>
using namespace std;
class WordSet
{
private:
    set<Word> set_1;
public:
    bool wordset_add(string s);
    void show();
};
```

WordSet.cpp

```
#include "WordSet.h"
bool WordSet::wordset_add(string s) {
    set_1.insert(Word(s));
    return 1;
}
void WordSet::show() {
    for (set<Word>::iterator it = set_1.begin(); it != set_1.end(); it++)
    {
        cout << (*it).getWord() << " ";
    }
}
```

创建WordMap 映射类

和前两个类一样。

WordMap.h

```
#pragma once
#include <iostream>
#include <map>
#include "Word.h"
class Wordmap
{
private:
    map<Word, int> map_1;
public:
    bool wordmap_add(string s);
    void show();
};
```

WordMap.cpp

```
#include "Wordmap.h"

bool Wordmap::wordmap_add(string s) {
    map<Word, int>::iterator it = map_1.find(s);
    if (it == map_1.end()) {
        pair<Word, int> p(Word(s), 1);
        map_1.insert(p);
    }
    else {
        it->second++;
    }
    return 1;
}

void Wordmap::show() {
    for (map<Word, int>::iterator it = map_1.begin(); it != map_1.end(); it++) {
        cout << it->first.getWord() << ":出现" << it->second << "次" << endl;
    }
}
```

mycode.h

```
#pragma once

#include "Word.h"
#include "Wordmap.h"
#include "WordSet.h"

#include <fstream>
#include <string>
#include <iterator>
#include <algorithm>
#include <numeric>
#include <sstream>

using namespace std;


void dataProcessing();
void showData();
```

mycode.cpp

```
#include "mycode.h"
//建立集合和映射的实例对象
WordSet set_111;
Wordmap map_111;
void showData() {
    cout << endl;
    cout << "单词集合为：" << endl;
    set_111.show();
    cout << endl;
    cout << "单词集合及出现次数：" << endl;
    map_111.show();
    cout << endl;
}
```

mycode.cpp

```
//打开文件读入数据，装入集合  
void dataProcessing() {
```

```
    //打开文件，读取文本  
    int pos = 0;  
    string s = " ";  
    string delimet = ",.?";  
    ifstream in1;  
    in1.open("data.txt");  
    char c;  
    while (in1.peek() != EOF)  
    {  
        in1.read(&c, 1);  
        cout << c;  
    }  
    in1.close();
```

```
        ifstream in("data.txt");  
        while (!in.eof())  
        {  
            getline(in, s);  
            if (s == " ") {  
                continue;  
            }  
            pos = 0;  
            while ((pos = s.find_first_of(delimet, pos)) != string::npos) {  
                s.replace(pos, 1, " ");  
            }  
            istringstream stringeam(s);  
            while (!stringeam.eof()) {  
                stringeam >> s;  
                if (s == " ") {  
                    continue;  
                }  
                set_111.wordset_add(s);  
                map_111.wordmap_add(s);  
            }  
        }  
        in.close();  
    }
```

test001.cpp

```
#include <iostream>
#include "mycode.h"
using namespace std;

int main() {
    dataProcessing();
    showData();
    return 0;
}
```

知识点总结

1. 案例展示
2. Word类
3. wordset类
4. wordmap类
5. mycode.cpp
6. main函数调用

作业

1. 自己动手完整实现该案例
2. 找出程序中的bug改正

bug提示，加上标点符号！在运行试试。